

Project dossier

Everything found, why it was pursued, what it produced, and what it means.

Saint-Gobain Research India · Varun Mandy, Research Intern [pvdlowe](#) v0.1.0 · 9,586 lines · 136 tests · 38 candidate architectures
<https://github.com/VarunMandy/pvdlowe>

How to read this

Every section answers four questions in the same order: **what** was done, **why** it was worth doing, **what** came out, and **what** it means. Sections are ordered by how much they matter, not by when they happened.

The single caveat that governs everything. No films were deposited. Every performance figure is a model output; median error against traceable literature is 30.6%. The largest known error — an eightfold under-prediction of sputtered copper sheet resistance — sits in the material most leading candidates use.

Parts II and III do not depend on any measurement. They are audits of the brief's method and evidence, and they stand alone.

PART I — THE ANSWER

1. What was asked

What. The brief proposed a materials-by-design programme for low-emissivity coatings on float glass: a dielectric/metal/dielectric stack, AZO/Ag/AZO, asking how much silver could be removed while holding visible transmittance, sheet resistance and far-infrared reflectance.

Why. Silver dominates both the cost and the supply risk of Low-E glazing. At roughly 0.105 g/m² for a 10 nm layer it is the single largest materials cost in the stack.

Result. The question turned out to be underspecified in a way that changes the answer, and the framework's most useful output was identifying that.

2. On the brief's own terms, almost no silver can be removed

What. Thin the silver in AZO/Ag/AZO; hold everything else.

Result.

Ag nm	T _{vis}	ε _h	R _s	Status
10.5	0.872	0.057	3.88	OK
10.0	0.876	0.060	4.18	OK — the limit
9.5	0.876	0.080	5.90	discontinuous

Percolation is the binding constraint, not any of the three specified properties. A 5% thickness reduction costs 41% in sheet resistance. Re-optimising both oxides at every thickness does not move it — the wall is in the metal.

Conclusion. The literal answer is **about half a nanometre**. That is a negative result and should be stated plainly. The 82–86% reduction reported below comes from changing the **composition**, not the thickness.

3. The answer depends on a question the brief did not ask

What. The brief specifies no climate, and the weighting does not constrain solar heat gain — which silently encodes a heating-dominated assumption, in a project run at Saint-Gobain Research **India**.

Result. Adding the EN 410 / ISO 9050 solar-gain metrics and re-scoring gives two rankings that **share one candidate out of ten**.

	Heating-dominated	Cooling-dominated (India)
Lead candidate	Si ₃ N ₄ /Ag ₁₀ Cu ₉₀ /Si ₃ N ₄ (E10e)	AZO/Cu/AZO (M6)
T _{vis}	0.877	0.738
Emissivity ε _h	0.051	0.046
Sheet resistance	3.13 Ω/sq	2.87 Ω/sq
Solar heat gain g	0.758	0.558
Silver	0.015 g/m ² (−86%)	zero
Meets spec	yes	no — T _{vis} 0.738 < 0.80

Benchmark: AZO/Ag/AZO at 10 nm — T_{vis} 0.876, ε_h 0.060, R_s 4.18, 0.105 g/m².

The mechanism is physical. AZO is a *transparent conductive oxide*: free carriers put a screened plasma edge at 1.25 μm, inside the solar band. Si₃N₄ is a passive insulator. Same 11 nm copper layer, transmittance at 1200/1600/2000 nm: AZO 0.240/0.115/0.068 against nitride 0.509/0.259/0.151.

Conclusion. A results table is meaningless without its weighting file. **The climate must be specified before any recommendation is made.**

4. The composition optimum is 5-15 at.% Ag, not 70%

What. Eight composition curves — Ag 0-100% in 5% steps, geometry re-optimised at every point, two microstructure models, two dielectrics, two climates.

Result. Every one peaks between 0 and 10 at.% Ag, and the segregated/AZO curve is flat from 5 to 40 at.% — a forgiving production window.

It is a sustainability result, not a performance one. Emissivity and sheet resistance keep improving up to 50 at.% Ag. The score falls anyway because silver mass carries weight: about 0.005 in emissivity for an 87% cut in silver.

Conclusion. Ag₇₀Cu₃₀ is outperformed in all eight combinations. **And the brief's own basis for it does not transfer** — it traces to a 2022 paper supporting 70/30 at 10 nm, but on **polycarbonate**, for first-surface coatings. Given \$9, that optimum does not carry to a glass line unchecked.

5. Two metal layers break a ceiling one cannot

Why it was tested. The best light-to-solar-gain ratio across all 38 single-metal candidates is **1.37**, against roughly 2.0 commercial. Under the cooling profile no single-metal stack met the transmittance target at all. That is an architectural limit, and no compositional search could have found it.

Result.

Architecture	Metal	Dielectric	Geometry (nm)	T _{vis}	g	LSG	ε _h	R _s
Single	Cu	AZO	35/12/35	0.738	0.558	1.32	0.046	2.87
Double	Ag	Si₃N₄	15/12/60/12/15	0.819	0.464	1.76	0.025	1.67
Double	Cu	Si ₃ N ₄	59/13/105/13/59	0.716	0.516	1.39	0.024	1.33

LSG 1.76 against a ceiling of 1.37 — 28% better. A single film trades transmittance against infrared reflectance along one curve; the dielectric *between* two metals adds an interference degree of freedom.

Note the optimum geometry: **thin outer, thick middle**. A symmetric guess misses it.

A third layer is not worth adding — LSG goes 1.33 → 1.76 → 1.81, and for copper it collapses transmittance to 0.437.

Conclusion. The architecture argument stops at two layers, and silver consumption doubles.

6. No candidate meets the Indian building code

What. Two criteria the brief left unspecified, anchored to ECBC 2017 rather than invented: SHGC ≤ 0.27, assembly U ≤ 3.0 W/m²K.

Basis warning. ECBC specifies *assembly* values. The framework computes centre-of-glass, single-pane. The conversion must be stated.

Result. Chaining a clear second pane (T_{sol} 0.83):

	g single	g DGU	vs 0.27
M6, the best	0.558	0.463	fails by +0.19
N35e, the worst	0.793	0.658	fails by +0.39

Not one of the 38 meets it, and the best misses by seventy per cent.

And the two targets are mutually incompatible. Across 990 double-metal designs, ECBC's 0.27 is reachable — at T_{vis} 0.644. The brief asks for 0.80. Since ECBC's own visible minimum is 0.27, the conflict is with the brief's **self-imposed** target, written for a heating climate.

Conclusion. External regulatory confirmation of \$5. For Indian glazing the single-metal architecture is not merely suboptimal — **it is non-compliant**, and the double-metal stack becomes a requirement.

PART II — THE AUDIT

7. Six defects in the proposed weighting

Why this matters more than it sounds. A scoring function is where every physical result becomes a ranking. A defect here corrupts everything downstream while leaving the physics untouched and every test passing.

Four in the table as proposed:

(a) Silver consumption carried zero weight. \$14 has no line for it, though \$17 and \$20 both name minimising silver as an objective.

The thickness optimiser exploited this immediately, returning a design using **29% more silver than the benchmark, at a perfect 100/100.**

(b) Emissivity and sheet resistance double-count at $r = 0.996$ — the same free-carrier response of the same layer, 0.40 of the total on one quantity.

(c) Targets at the specification minimum. Setting T_{vis} 's target to 0.80 — the brief's *floor* — made the criterion flat where candidates differ. Across 181 oxide pairs clearing 0.80, transmittance spanned 0.800–0.881 while the **score went down**.

(d) The supply-risk band could not discriminate — every silver-bearing composition scored 0.1–0.28.

Two more, found by re-auditing the fixes for (a) and (b):

(e) A triple-count. Silver mass, metal cost and supply risk correlate at $r = 1.000$ and **0.991** — cost is silver in different units, and supply risk in this set *is* silver's. Together **36% of the effective weight** on one property.

(f) Weight on criteria never populated. Three criteria carried **0.30 of a 0.90 nominal total while being None** for every candidate, renormalising away silently — a file stating emissivity at 0.25 was applying 0.42. One of them had **no criterion definition at all**.

And a seventh issue: aggregation. The brief asks its weighting to prevent a candidate "winning simply because it has excellent conductivity while being unacceptable optically." A weighted sum cannot; a geometric mean can.

A residual that cannot be corrected, only disclosed. The silver weight of 0.15 is a judgement and it selects the answer — the winner is a 0.065 g/m² design at weight 0 and silver-free at 0.30, changing three times in between. **Five of eight settings separate first from second by under half a point.**

`weight_sweep()` reports the transitions instead of a single ranking. **The honest recommendation is E10e at 0.15, E5e at 0.20–0.25, and the silver-free N6 at 0.30 and above — and choosing among them is a decision about how much silver consumption matters to Saint-Gobain, not one the framework can make.**

Conclusion. Four corrected, two found and corrected on re-audit, one residual disclosed. **General rules: a desirability target should be an aspiration, not a specification minimum; and a weight that changes the winner should be reported as a sweep, not a value.**

8. The evidence base: eight of fourteen claims changed

Why. Every citation carried `utm_source=chatgpt.com` — LLM-surfaced rather than read from a publisher. Not necessarily wrong, but unchecked.

Status	Count
Verified — full text read	3
Partial — located, confirmed against abstract	4
Disputed — appear in no locatable source	2
Supported — corroborated independently	3
Not located after searching	1

Two citations cannot be supported and should not be cited.

An uncomfortable corollary. These two are the model's three *closest* agreements — 0.3%, 1.0%, 1.2% relative error. Excluding them, the median validation error rises from **14.7% to 30.6%**. `validate_model()` now excludes them by default, and a test fails if that exclusion ever starts *lowering* the reported error.

One transcription question resolved the other way. The brief quotes 1.59 $\mu\Omega\text{-cm}$ for a 10 nm Ag film. The journal states 1.59 — **the brief is faithful** — while the group's own patent states 1.29, below bulk silver and impossible. Both are anomalous by a factor of three: Fuchs-Sondheimer gives 4.74, this framework 4.44, an independent 12 nm measurement 4.8.

9. A measurement that contradicts a model-independent limit

What. *Applied Surface Science* **578** (2022) reports AZO/Cu/AZO with T_{vis} 87.7%, R_s 9.96 Ω/sq and ϵ 0.055. **The brief's §16 conclusion rests entirely on it.**

Why it can be judged without any model. For a sheet much thinner than the wavelength, far-infrared reflectance is fixed by sheet resistance alone through the impedance of free space. At 9.96 Ω/sq the floor is $\epsilon \geq \mathbf{0.096}$.

Six explanations tested and eliminated:

Explanation	Outcome
Different samples	Abstract attributes all three to one sample
Transcription	Confirmed against the abstract
Far-IR glass index	Limit is 0.091–0.097 for $n = 1.5\text{--}4.0$
Band-limited emissometer	Reads 5–8% higher — wrong direction
The framework's own convention	An industrial formula since 2000 agrees
Measurement basis	Hemispherical fails by 51% rather than 42%

And the two copper measurements sit in opposite positions:

	Technique	R_s	ϵ	Floor	
Miao 2014, HK PolyU	RF	16.60	0.330	0.150	consistent
ApSS 2022, Xi'an	DC	9.96	0.055	0.096	below floor

The measurement that obeys the limit is the one that contradicts this framework, by a factor of 7.7. The one that agrees with the model's optimism is impossible. That contrast is stronger than either alone.

Conclusion. §16's conclusion should not be used until settled.

PART III — WHERE THE MODEL WAS WRONG

10. The dielectric finding was reversed by measurement

What. The framework concluded Si_3N_4 beats AZO on transmittance by up to nine points, on index matching. Cueva & Carretero (*Coatings* **13**, 1709, 2023, open access, **full text read**) deposited five dielectrics under identical conditions.

Result — the opposite. AZO wins on emissivity (0.058) and transmittance, despite the lowest refractive index of the five. The authors' reason: silver grows more efficiently on AZO.

What was missing. The model treated the dielectric purely as an interference layer. The optical inputs were sound. What was absent is that **the underlayer determines the quality of the metal grown on it.**

The correction. `metal_growth_factor` per underlayer, calibrated to that series. The model now reproduces the measured ordering to 4-8%.

The productive outcome is a hybrid — AZO beneath for nucleation, nitride above for durability — which is what industry already does:

ID	Stack	T_vis	ε_h	Heating
H1	AZO/Ag/Si ₃ N ₄	0.918	0.057	69.6
M0	AZO/Ag/AZO (benchmark)	0.876	0.060	66.6
H2	AZO/Cu/Si ₃ N ₄ (silver-free)	0.788	0.042	77.6

H2 is the only architecture in both climate top-tens.

11. The mechanism, and the principal structural weakness

What. Two ML surrogate calculations were run to find the mechanism behind `metal_growth_factor`. **Both failed**, each documented with the diagnostic that caught it:

Attempt	Failed on	Diagnostic
Bulk adhesion	5–21% lattice mismatch — measured elastic strain	unphysical 9.6 and −0.16 J/m ²
Adatom wetting	termination changed the answer by 2.1 eV against 0.46 eV between materials	explicit termination sweep

A literature search found it in under an hour. US 7,632,572 B2 (AFG → AGC → Cardinal CG; **full text read**) deposited 16 nm Ag on amorphous TiOx and on a 5 nm ZnO seed over the same TiOx:

- **25 nm grains on ZnO against 15 nm on a-TiOx**
- {111}-oriented grains **two to three times larger** on ZnO
- the film on bare amorphous titania **clearly discontinuous**

The inventors name the mechanism: ZnO grows {0001}, orienting the silver to grow {111}, and the epitaxial lattice match lowers sheet resistance and improves adhesion.

An unexpected validation. The patent reports four-point sheet resistance — **5.68 Ω/□ with the ZnO seed against 7.56 without**, ratio **1.331**. `metal_growth_factor`, calibrated independently from a different group's *emissivity* series, gives **1.250**. **Agreement to 6%** — a parameter fitted to one dataset reproducing another it never saw.

The most consequential statement in this work. The framework models each metal layer as though the layer beneath it did not shape its microstructure. That single omission accounts for **both** `metal_growth_factor` **and** the eightfold copper under-prediction. They are one effect appearing twice.

12. A seed layer may account for most of the copper discrepancy

Result. Six further AZO/Cu/AZO studies located:

Film	Seed	R_s
AZO/Ti/Cu/AZO, <i>Opt. Lett.</i> 2017	1-2 nm Ti	4.31 Ω/sq
AZO/Cu/AZO, ApSS 2022	none	9.96
AZO/Cu/AZO, Miao 2014	none	16.6
this framework, 15 nm Cu	—	2.16

The framework is eightfold optimistic against unseeded copper and roughly twofold against seeded copper. So the model is not wrong about copper as a material — **it is wrong about poorly nucleated copper.**

Same mechanism as §11, now visible in three independent places.

Conclusion. The thickness series should include a seeded arm. One extra target converts the largest known error from an anomaly into a process variable.

13. Corrections to the project's own conclusions

Five conclusions stated during this work were overturned **by the work itself**, and are recorded rather than quietly fixed.

Claim	Overtured by
An optimum exists at Ag ₆₀ Cu ₄₀	A three-point artifact; at 5% resolution the curve is monotonic
The Ti barrier variant beats the ternary	An artifact of silver carrying zero weight
A band-limited emissometer explains the ApSS emissivity	Tested: it reads 5-8% <i>higher</i>
Adatom wetting reproduces the measured ordering	A termination sweep showed it resolved slab index
The brief mistranscribed a resistivity	The journal says 1.59; two publications disagree

Three of five are the same error: reading structure into too few points.

Two are a different error, and it recurred: holding a fragment of a source and treating it as the source. The project's own [LITERATURE](#) versus [LITERATURE_UNVERIFIED](#) grading exists to prevent exactly this and was being applied to data but not to reasoning.

PART IV — WHAT WAS COMPUTED

14. Ag-Cu has no stable ordered compound

All fifteen nominated systems screened against the Materials Project convex hull. Two intermediate Ag-Cu phases exist and **neither is stable**: Cu₃Ag at **0.0904 eV/atom**, CuAg₃ at **0.0857** — well above k_{BT} at deposition.

A second finding. Cu-Ti has 8 near-hull phases and Al-Cu has 10, so a dilute Ti or Al addition has stable intermetallics available to precipitate rather than remaining in solution — a second mechanism for the ternary underperforming.

15. The dilute corner is thermodynamically favoured

MACE-MP-0 and CHGNet on 32-site fcc supercells, **gated against the two known hull distances before any prediction was accepted**. Positive across the whole range: $\Delta E_{mix} = 0.287 \cdot x \cdot (1-x)$, fitting to ± 6 meV/atom.

A self-check that matters more than the gate. At Ag 25 at.% the disordered solution sits 13 meV/atom **below** ordered Cu₃Ag — no ordering tendency at all, exactly what an empty hull implies, reached independently.

Ag at.%	ΔE_{mix}	$\times kT$ (550 K)
70 — the brief's priority	0.0602	1.27
15	0.0366	0.77
10	0.0258	0.54
5	0.0136	0.29

In the dilute corner the driving force to separate falls below thermal energy at deposition. §4 found the microstructure hypotheses converge there; the thermodynamics supply the reason.

The dilute optimum is supported three independent ways: least silver, best corrected score, weakest tendency to phase-separate.

Cross-checked, with a caveat. Both models under-predict, so the bias is inherited from shared training data. **They are not independent** — agreement shows the bias is consistent, not absent. And GGA itself underestimates Cu-transition-metal formation energies by ~40%, so two errors act in series. The conclusion **holds at 5-10 at.% Ag, marginal at 15%**.

16. What a surrogate cannot do

Cannot, at all: optical constants, dielectric functions, band gaps, emissivity. These models carry no electronic structure.

And a conductivity result does not transfer to an optical one. The emissivity-sheet resistance link assumes a low-damping Drude metal, a property of *crystalline* silver. A conductivity gain from amorphisation cannot be assumed to give the corresponding emissivity gain.

PART V — WHAT REMAINS

17. The critical unknown

The framework validates against silver to within 0.4% and **fails on copper** by roughly eightfold. **Six of the leading candidates are copper-based.**

The classical size effect is eliminated. An exhaustive scan over the entire admissible range of published scattering parameters reaches only **4.42 Ω/sq against the 16.6 to be explained** — short by a factor of 3.8. Grain structure is the leading hypothesis; §12 adds that a seed layer recovers most of the gap.

If that reading is right, copper's obstacle is film quality, not physics — an engineering problem with known levers.

18. The experiment that resolves it

One XRD scan, one film, an afternoon — three questions answered.

Measurement	Question
Peak count and position	Microstructure — segregated gives two fcc sets, a solid solution one
Scherrer width	Grain size , and therefore grain_size_ratio
(111)/(200) vs the powder value of 2.2	Templating , the mechanism of §11

A second, independent discriminator. Sheet resistance separates the hypotheses by a number; diffraction by *peak count*. Two unrelated observables agreeing is far stronger than either — and both come from one film.

Four requirements, all cheap: scan to $2\theta = 100^\circ$ not 60; grazing incidence rather than Bragg-Brentano, because a copper peak here is marginal and a null would otherwise be ambiguous; a 30 nm uncapped calibration film; and a seeded arm.

Pre-registered decision rules — four outcomes, each with what it means and what to do, **including the one where six candidates fall over.**

19. The questions only SGRI can answer

Which stack is the right baseline. This work benchmarks against 10 nm Ag at ϵ_h 0.060, from the brief — whose citations for it are the two **disputed** entries. Saint-Gobain's own patents claim **12.5-16 nm targeting $\epsilon \leq 0.038$** , which the model says needs 14-16 nm. **If production runs there, the reported reductions are measured against a stack that would not meet SGRI's own spec.**

Which transmittance target is real. The brief's 0.80, or ECBC compliance at 0.644. §6 shows they cannot both be met.

PART VI — WHAT WAS BUILT

20. The framework

9,586 lines, 136 tests, twelve subpackages, fourteen CLI commands.

The design decision that matters. Film resistivity feeds the Drude damping, so a metal layer too thin to conduct is automatically one that reflects poorly. Without that coupling, optimising for transmittance drives silver thickness to zero and reports an excellent coating.

Validated against closed-form answers, not by inspection: exact Fresnel at a bare interface, an exact null for a quarter-wave layer, energy conservation to 10^{-10} , s and p agreeing at normal incidence.

Provenance as a type. Eleven evidence grades, and a guard that refuses hypothesis-grade values in headline tables. It is what kept the Ti-ternary predictions out of the results.

Functions that refuse rather than guess. The sputter model raises when uncalibrated; the MP client raises when offline with a cold cache; the surrogate refuses to fall back to an empirical potential; the adhesion calculation refuses above 4% lattice mismatch. **Each was tempting to make helpful; each would have produced fabricated data.**

21. Known limitations

No experimental validation. Median error 30.6% against traceable literature.

The principal structural weakness, restated because it governs the rest: the framework models each metal layer as though the layer beneath it did not shape its microstructure.

Fitted parameters. Specularity and grain-boundary reflection are fitted; Si_3N_4 and TiO_2 optical constants are **ESTIMATE** grade.

Illuminant approximation, quantified. T_{vis} moves 0.1% over 5000–7500 K; T_{sol} moves 8.3%. T_{sol} , g and LSG carry a 5–10% systematic.

Scores are not portable. They moved ~25 points as the \$7 defects were fixed. **A score is not a property of a material.**

Not modelled at all: roughness, interdiffusion, damp-heat and abrasion durability, adhesion, agglomeration kinetics, amorphous metal layers.

CONCLUSION

What this project established. A better composition — 5–15 at.% Ag rather than 70%, on three independent grounds. A better material class — silicon nitride, outside the brief's framing, with the hybrid arrangement industry already uses. A better architecture — two metal layers reaching LSG 1.76 against a ceiling of 1.37. And a specified climate dependency, with the two profiles sharing one candidate out of ten.

What it corrected. Six defects in the proposed weighting, one of which let the optimiser return **29% more silver than the incumbent at a perfect score**. Two citations that cannot be traced. One measurement inconsistent with a model-independent limit, six explanations eliminated. And five of its own conclusions, overturned and recorded rather than hidden.

What it cannot claim. Nothing has been deposited. The rankings are hypotheses ordered by physics. Their principal weakness is known, quantified, and addressed by a specified experiment.

The recommendation. One XRD scan, one film, an afternoon.

And two questions to Saint-Gobain: which stack is the right baseline, and which transmittance target is real.

A model that identifies the measurement capable of falsifying it is more useful to an experimental programme than one that does not. That is what this framework is for, and it is what it delivered.